Since each of the above equations represents a straight line, they must be of the first degree in x and y. Therefore the expression under the radical sign should be a perfect square and the condition for this is

$$4\left(gh - af\right)^2 - 4\left(h^2 - ab\right)\left(g^2 - ac\right) = 0$$

Simplifying and dividing by a,

We get, $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ or

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$ (Expansion of Determinant will be studied in the next chapter)

Results without Proof:

(i) Two straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ are

parallel if $\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$ or $bg^2 = af^2$

(ii) If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of parallel straight lines, then

the distance between them is $2\sqrt{\frac{(g^2 - ac)}{a(a+b)}}$ or $2\sqrt{\frac{(f^2 - bc)}{b(a+b)}}$

The relationship between the equations of pair of straight lines

$$ax^2 + 2hxy + by^2 = 0 \quad (6.41)$$

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0) \quad (6.42)$$

The slope of the above pair of straight lines (6.41) and (6.42) are depending only on the coefficients of

x^2, xy and y^2.

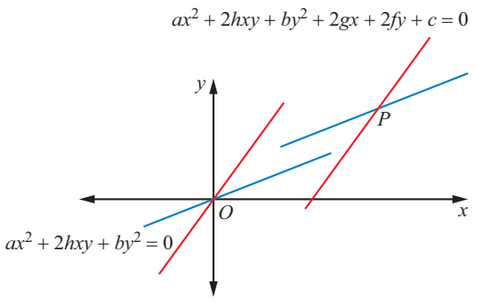


Figure 6.51

(i) If the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines, then $ax^2 + 2hxy + by^2 = 0$ represents a pair of straight lines through the origin parallel to the first pair. The point of intersection (6.41) is (0, 0) and the point of intersection (6.42) is

$$P\left(\frac{hf - bg}{ab - h^2}, \quad \frac{gh - af}{ab - h^2}\right)$$

(ii) If θ be the angle between the straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, then it will have the same value as the angle between the two lines represented by $ax^2 + 2hxy + by^2 = 0$

$$\text{Thus } \theta = \tan^{-1}\left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$$

(iii) If the two straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ are at right angles, then the two lines represented by $ax^2 + 2hxy + by^2 = 0$ are also at right angles and the condition is $a + b = 0$